PIGNN-UQ: Physics-Informed Graph Neural Networks with Uncertainty Quantification ~~: A Pre-Registered Evaluation Protocol~~ for Registry-Compatible Blue Carbon ~~Credit Pricing~~ Crediting

Nahla Nabil Skaik
Faculty of Computer Studies
Arab Open University
Kingdom of Bahrain
Email: nahla.nabil.52@gmail.com

***Abstract*—~~Flat, rule-based~~ Mangrove carbon credits are discounted by a non-permanence ~~buffers discount every credit in a voluntary carbon market identically, regardless of~~ buffer that does not depend on how uncertain the underlying carbon-stock estimate ~~'s confidence. We test whether~~ is. In this study, PIGNN-UQ combines a physics-informed graph neural network ~~(GNN) with conformalized quantile regression (CQR) can supply a per-site confidence signal that complements, rather than replaces, this buffer, on a 10,124-node spatial graph of the Ten Thousand Islands mangrove fringe, southwest Florida (Sentinel-2, Global Mangrove Watch, field-calibrated~~ with gradient-boosted trees to map soil organic carbon ~~). Model selection (physics-loss weight,feature set,calibration method)was pre-registered on validation data and confirmed once on a~~ over two southwest Florida mangrove regions (10,124 and 22,326 nodes of 150 m), and conformal prediction turns each estimate into a 90% interval, calibrated on held-out ~~test block. A five-seed check found~~ global-map labels, that feeds the uncertainty deduction of the Verra VM0033 methodology. The mean of the physics-informed ~~loss's point-accuracy advantage over a no-physics GNN did not replicate: on test , the no-physics GNN has the best~~ network and XGBoost, selected by spatial cross-validation, reached a test R^2 of ~~three models (−0.383 vs. −0.430 physics-informed and −0.450 linear). Its predictions do correlate most strongly with the imposed physical prior on held-out nodes (0.755 vs. 0.682/0.686), evidence the constraint transfers under spatial shift, not independent physical validation, and a real but much smaller margin than an earlier single-seed pilot suggested. Global CQR calibration reached 91.1% /96.4% coverage at 90%/~~0.28 and 0.60 (declared second test read), against 0.20 and 0.59 for XGBoost. The intervals covered 91.5% and 89.1% of test nodes, and their width tracked absolute error (Spearman 0.54 in the first region). The registry rule withheld a median 12.7% and 2.1% of credits per node, of the same order as the 11.0% median buffer of the seven issuing blue carbon projects. The label map's own uncalibrated intervals would withhold about 65%. The methodology's independent-error aggregation covered only 18% and 33% of realized project-level errors (upper 95~~% nominal; a spatially localized (Mondrian) variant undercovered more on test (83.4% , 92.4%), a genuine negative result consistent with spatial-conformal theory's local-density requirement. A pricing proof-of-concept built on the better-calibrated global intervals maps interval width to a multiplier ranging 0.50–1.50 across five test nodes, against Plan Vivo's fixed 0.800 flat-buffer multiplier. Point-prediction accuracy remains weak throughout; the contribution is the pre-registered protocol itself, which caught both a single-run estimate and a calibration refinement overstating their reliability, not validated point accuracy or a validated pricing model.~~ % bounds 52% and 49%). Empirical project-level calibration covered 91%. Against 87 soil cores at 25 sites, the ensemble had the lowest median error (10.1%, against 14.1% to 15.1% for two published maps and a regional mean), though within sampling uncertainty. These results motivate registry-scale tests of crediting rules that keep estimation uncertainty separate from non-permanence risk and aggregate it under spatially correlated errors.

***Index Terms*—blue carbon, graph neural ~~network, physics-informed learning,~~ networks, conformal prediction, ~~uncertainty quantification, carbon credit pricing~~ carbon credits, uncertainty quantification**

I. Introduction

~~Voluntary carbon markets discount mangrove credits for~~ Under Verra's VM0033 methodology, a mangrove project deposits part of its credits in a pooled buffer account against non-permanence risk (carbon later released ~~via~~ through erosion, sea-level rise , or disturbance)~~using a buffer pool: a fixed percentage of issued credits withheld from sale, unconditioned on how confident the underlying carbon-stock estimate is. Verra's VM0033 and Plan Vivo V5.0 both apply this uniformly, regardless of sub-project granularity [1], and existing carbon-stock models for this ecosystem and region report only point estimates, with no calibrated per-prediction confidence a pricing rule could condition on [2]. Conformal prediction supplies valid, distribution-free uncertainty for earth-observation and spatial regression [3], [4], but no existing work pairs that calibrated uncertainty with a physics-informed spatial predictor, couples it to a buffer-pool pricing mechanism, or tests whether a calibration strategy (global versus spatially localized) holds under one region's calibration-density constraints, the narrower gap this paper addresses (Section II surveys each literature in full). This paper~~ . A separate deduction applies when the 90% confidence interval of the estimate exceeds 20% of its value [1, p. 64]. The buffer share comes from a risk assessment and does not depend on how well the carbon stock itself was estimated. Published mangrove carbon maps either report no per-site interval or report intervals whose coverage and crediting effect have not been checked [2], [5]

, and the methodology combines plot uncertainties under an independence assumption that spatial models rarely satisfy.

This paper asks whether calibrated, per-site uncertainty from a spatial model can supply that deduction in a way a registry could audit. The proposed pipeline, PIGNN-UQ (Fig. 1), combines a physics-informed graph neural network ~~[6], previously applied to mass-conserving flood forecasting [7] but not, to our knowledge, blue carbon, with distribution-free uncertainty calibration to test whether estimation confidence can complement, not replace, a flat~~ (GNN) that encodes neighborhood structure and a tidal-inundation prior, tree ensembles as strong tabular baselines, and conformalized quantile regression for intervals calibrated on held-out data. This study departs from earlier work in four ways: (1) the interval of every node is converted into the VM0033 deduction, kept separate from the non-permanence ~~discount.~~ buffer and compared with the buffers held by issuing projects; (2) the methodology's own aggregation rule is tested empirically and shown to understate project-level uncertainty when model errors are spatially correlated; (3) predictions and published maps are checked against 87 independent field soil cores; and (4) the analysis covers two regions, two image dates and leakage-free spatial splits, with model choices fixed on validation data under rules committed in advance.

The scope is deliberately applied: the paper does not propose a new network architecture. The contribution is a crediting pipeline whose uncertainty is calibrated, auditable and conservative, together with evidence on where that uncertainty is and is not informative. Table I positions this contribution against prior work.

~~The scope is deliberately narrow: one pilot region, one physics constraint (a steady-state mass-balance proxy, not a full temporal flux equation), and one primary UQ method (CQR; deep ensembles and MC dropout are left for future work). Every model-selection decision, physics-loss weight, feature set, calibration method, was pre-registered on a validation split and confirmed once on a spatially distinct test block never used for any prior decision; we treat this two-stage protocol itself as a methodological contribution (Section III-C) , since a naive single-run estimate and a theoretically-motivated calibration refinement each turned out to mislead here (Section V) . What survives that scrutiny is smaller and more tightly bounded than either initial result suggested: a real but modest prior-generalization advantage for the physics-informed model, and evidence that localized (Mondrian) conformal calibration, plausible in theory and competitive on validation-internal evidence, underperforms the simpler global variant on this AOI's test data . The contribution is this pre-registered evaluation and its honest limits, not a validated point-prediction accuracy claim or a validated pricing model.~~

II. Related Work

Physics-informed and graph learning. Physics-guided ~~ML integrates scientific structure into learned models [8]; PINNs add a PDE-residual loss [6], Graph Network Simulators encode physics via local, permutation-invariant message passing . The closest prior art, flood forecasting [7], is entirely hydrological and does not target mangrove carbon or couple prediction to a priced instrument via UQ~~learning adds scientific structure to data-driven models [8]. Physics-informed networks penalize violations of governing equations [6], and graph formulations have carried such constraints into flood forecasting [7]. GraphSAGE [9] learns from node neighborhoods inductively, which suits spatial grids split into separate regions.

TABLE I
Requirements of model-based crediting covered by prior work

Strand of work	Interval	Rule	Aggregation	Field	Market
Mangrove carbon maps	○	×	×	✓	×
Conformal prediction in EO	✓	×	×	×	×
Registry methodologies	×	✓	×	✓	×
Blue carbon market studies	×	×	×	×	✓
This study	✓	✓	✓	✓	✓

Interval: calibrated per-site interval (○: reported, not calibrated); Rule: registry deduction; Aggregation: tested on realized errors; Field, Market: checked against cores, registry records.

~~The dominant approach is per-pixel tabular regression (random forest, XGBoost) on spectral indices. Simard et al.'s global mangrove biomass product was inaccessible here (NASA Earthdata login required). Radabaugh et al. built a field-plot-calibrated, 30 m carbon-stock model for~~ **Mangrove carbon mapping.** Mangrove carbon studies mostly fit linear models to remote-sensing predictors, with random forests and gradient boosting increasingly common [10]. Global soil carbon products are themselves machine-learning outputs [11], whereas regional maps such as the southwest Florida ~~coast, one sub-region matching our pilot AOI [2]. Sanderman~~ map of Radabaugh et al. ~~produced the global, field-calibrated 30 m SOC product we train against. A 105-study review confirms belowground-carbon and valuation research remain under-studied versus aboveground biomass [10] . None uses a graph representation or a physical constraint, or reports a calibrated uncertainty interval (the gap this paper addresses)~~are calibrated on local field cores. Geomorphic setting and tidal regime control mangrove soil carbon [12], [13], and soil carbon maps can depart substantially from core data [14]. Random cross-validation overstates accuracy on such data, so spatial blocking is required [15], [16].

Uncertainty quantification. ~~MC dropout is cheap but poorly calibrated under shift; deep ensembles calibrate better but need multiple full models, as in climate-downscaling . CQR [17] turns a point predictor into calibrated intervals with a coverage guarantee that holds without distributional assumptions, so long as calibration and test data are exchangeable, validated for~~ Split conformal prediction gives finite-sample coverage under exchangeability [18], and conformalized quantile regression adapts the interval width to each input [17] and has been applied to earth observation [3]~~; that guarantee is conditional on exchangeabilityspatial autocorrelation can violate. GeoConformal Prediction [4] addresses this with a spatially localized framework, motivating the Mondrian variant we test (Section III-F)~~ . Spatial dependence weakens exchangeability, which motivates localized (Mondrian) calibration [4], [19], whose validity needs dense calibration data in each region [20]. Interval quality is scored with the Winkler interval score [21], [22].

Crediting and markets. Blue carbon credits traded at a weighted average of US$26.96 per credit in 2022 and 2023 [23], and registry records report issuances, retirements and buffer deposits per project [24].

Research gap. Table I compares these strands against five requirements of model-based crediting. No study, to our

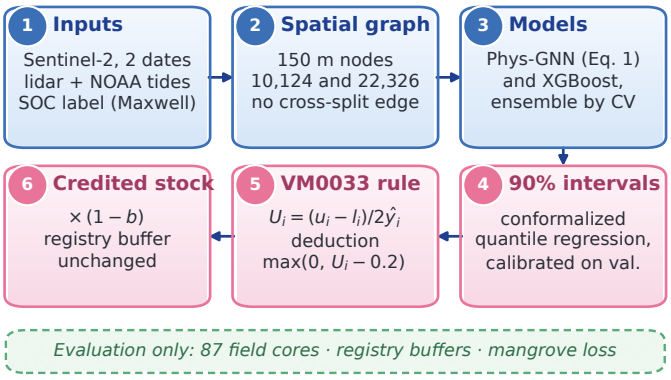


Fig. 1. PIGNN-UQ crediting pipeline. Field cores, registry records and mangrove loss are used only for evaluation.

knowledge, (i) converts calibrated spatial intervals into a registry uncertainty deduction, (ii) tests the registry's aggregation rule against realized errors, and (iii) checks the result against field cores and market records at the same time.

III. Data and Methods

~~The pilot AOI is the~~

A. Regions, data and graph

Two southwest Florida mangrove regions were used: Ten Thousand Islands / Rookery Bay ~~mangrove fringe, southwest Florida ((~~R1; −81.90° to −81.40°~~lon.~~, 25.90° to 26.30°~~lat.) , chosen for free access to every needed layer: Sentinel-2 L2A via Microsoft Planetary Computer, Global Mangrove Watch (GMW)v3 extent [25], the Sanderman~~) and the Everglades National Park coast (R2; −81.20° to −80.90°, 25.30° to 25.60°). The label is the 30 m soil organic carbon stock (0 to 100 cm, t C/ha) of the 2020 global mangrove product of Maxwell et al.~~ SOC product , and a NOAA CO-OPS tide gauge (station 8725114). Field-calibrated validation numbers came from Radabaugh et al. [2].~~

~~Native 30 m SOC grid cells were~~ , which also provides per-pixel 95% prediction limits. Native cells are aggregated into 5 × 5 blocks (~~~~about 150 m~~~~nodes) ; Sentinel-2 bands were resampled onto this grid (bilinear) and the GMW mask (nearest-neighbor), so the label was never resampled. A node was kept if its block-mean GMW mangrove fraction exceeded~~) kept where the Global Mangrove Watch 2020 fraction exceeds 0.5 ~~and it had a valid SOC value~~ [25] , giving 10~~,~~124 nodes ~~and (~~64~~,~~366 directed ~~edges (8-connectivity)~~ 8-neighbor edges) in R1 and 22,326 nodes (143,034 edges) in R2. Node features ~~were NDVI, raw B04/B08/B11 reflectance,~~ are ten Sentinel-2 L2A bands and derived indices [26] for 6 February 2026 and, with cloud masking, for 15 August 2026, together with elevation and distance to the tide gauge~~; the label was mean SOC (t~~. While revising this work, the Sentinel-2 clip of the earlier version was found shifted 9.18 ~~C/ha, 0–100 cm)~~ km north (mean normalized difference vegetation index (NDVI) of mangrove nodes 0.003). The shift was reproduced on correctly referenced imagery (correlation 0.91 to 0.97), and all features were rebuilt (mean NDVI 0.46). ~~An initial contiguous-band split gave negative test R^2 for every model family, since the held-out band straddled a real regional carbon-density gradient (Radabaugh et al.'s sub-regional figures range 265.1 ± 43.2 to 409 ± 104.4~~

Splits are spatial: fine cells of about 2.5 ~~Mg C ha^{-1} [2]). We replaced this with an 8 × 8 checkerboard grid of spatial superblocks (~~~~km~~ are assigned whole to training, validation or test (R1: 6 ~~km each, far larger than the 150 m node spacing), assigned by cumulative node count rather than superblock count, since mangrove nodes concentrate in a thin coastal fringe.~~

~~An early version of this evaluation used one fixed split, directly comparing test-set metrics across model variants to decide the paper's central claim, using test data beyond its role as a single untouched confirmation. We corrected this before any experiment below: a fresh validation /test block was carved from superblocks never individually inspected for a decision (only touched via gradient updates); the previously-inspected split was folded into the new training pool, since training on previously-seen data does not leak, but evaluating on it again would. The corrected split is 6~~,465~~train~~ /1~~,~~583~~validation~~ /2~~,~~076 ~~test nodes(label means 427.9/421.2/399.4 t/ha,AOI mean 421.0~~nodes; R2: 14,499/3,351/4,476). Every ~~later selection decision (physics-loss weight, feature set, calibration method) was made on validation only, by rules fixed beforehand, confirmed once on test (Section IV): a pre-registered "tested X, selected Y" claim this protocol makes defensible, surfacing a genuine instability a fixed split could not reveal~~model is trained and evaluated on subgraphs induced by one split, so no edge crosses a split boundary.

~~All models shared a 2-layer GraphSAGE backbone [9] (hidden dim.~~

B. Models and physics prior

Baselines are linear regression, a location-only random forest, random forest with and without neighbor-mean (spatial lag) features, and XGBoost [27], [28]. The graph model is a two-layer GraphSAGE network (hidden size 16, dropout 0.3~~), Adam(lr = 0.01, weight decay 10^{-3})~~, Adam, learning rate 0.01, early stopping on ~~validation MSE (patience 30). Baseline and~~ held-out groups). The physics-informed ~~variants differed only in loss function, isolating the physics term's effect. The full mass-balance relationship is dC/dt = Accretion − Export − Decomposition; a single Sentinel-2 date and no flux-tower data ruled out estimating this directly. We imposed a steady-state priorinstead: predicted stock was softly encouraged toward an equal-weighted, min-max-normalized (training data only) combination of an accretion proxy (NDVI) and an export proxy (tide-gauge distance, a positional stand-in for tidal energy absent a per-node hydrodynamic layer) . Both proxy components are also raw model features (Section III-B) ; withholding both from a no-physics baseline reduces its test prior-generalization correlation only modestly (0.682 to 0.654), so this is not narrowly a feature-overlap artifact. A second term penalized negative predictions, a mass-balance corollary, not independent. Both combined into one physics loss, added to the standard MSE data loss, with no fitted coefficients inside the proxy itself:~~ variant adds a steady-state prior,

$$\mathcal{L} = \mathrm{MSE}(\hat{y}, y) + \lambda\,\mathrm{MSE}(\hat{y}_z, P_z) + w\,\overline{\mathrm{ReLU}(-\hat{y})^2}, \quad (1)$$

~~$w_{\text{nonneg}} = 0.1$ was held fixed (never swept); λ was chosen by a pre-registered sweep over $\{0.01, 0.05, 0.1, 0.3, 0.5, 1.0, 2.0\}$, five seeds each, on validation only, selecting the value with best mean validation prior-generalization correlation within 0.05 of the best mean validation R^2. This rule selected~~ where $\hat{y}_z$ and P_z are the standardized prediction and prior, $\lambda = 0.1$ (~~mean val. $R^2 = -0.007$, mean val. prior-gen. corr. = 0.670),~~

not the ad hoc $\lambda = 0.3$ from an earlier pilot pass, which the rule did not select (full 7-value table in the project's number-source log).

A quantile-head GraphSAGE (same backbone) was trained with pinball loss on the 2.5th, 5th, 95th, and 97.5th percentiles, data loss only (no physics loss; combining both was future work). Two feature sets, NDWI (McFeeters, Green/NIR) and NDMI (Gao, NIR/SWIR), were tested as additions to base, five seeds each, on validation only, under the same tolerance rule as λ; neither improved mean validation prior-gen. correlation within tolerance, so base was retained (Section IV-A).

0.3 and 1) and $w = 0.1$ weight the prior and non-negativity terms. The prior P is the mean of min-max scaled tidal inundation frequency (the share of 2025 hourly National Oceanic and Atmospheric Administration (NOAA) water levels above each node's U.S. Geological Survey (USGS) 3DEP lidar elevation, both in NAVD88) and red-edge canopy index (NDRE), reflecting the role of hydroperiod in sediment delivery and carbon preservation. The prior acts as a soft domain constraint rather than a governing equation, and it replaced the NDVI and gauge-distance proxy of the earlier version. The final candidate set added the unweighted mean of the physics-informed GNN and XGBoost (the ensemble), a stacked variant in which the GNN corrects out-of-fold XGBoost predictions, and a GNN trained on both image dates. Table II lists every candidate with its accuracy.

Split-conformal calibration used the validation set: the conformity score $E_i = \max(q_{\text{lo}}(x_i) - y_i,\ y_i - q_{\text{hi}}(x_i))$ was computed there,

C. Uncertainty and crediting

Quantile models (GNN, physics-informed GNN, XGBoost), normalized conformal random forest and a quantile ensemble were calibrated by split conformal prediction on the validation split, either globally or per spatial block (Mondrian, at least 30 calibration nodes per block). The trivial interval is the training mean $\pm$ the conformal quantile of $|y - \bar{y}|$ on the same calibration nodes, and deductions are clipped to [0, 1]. For node i with estimate $\hat{y}_i$ and 90% interval $[l_i, u_i]$, the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $\{E_i\}$ adjusted the interval bounds (*global* calibration, one correction for every node). A *localized (Mondrian)* variant instead partitioned validation nodes by spatial superblock (Section III-B) relative uncertainty $U_i = (u_i - l_i)/(2\hat{y}_i)$ enters the VM0033 rule, computing a separate correction per block ($\geq$30 points, else global fallback). Both, plus a density-weighted variant (Gaussian kernel, bandwidth = median nearest-neighbor distance) , were compared on a validation-internal holdout (val split 50/50 into calibration/check, never reading test ; Section IV-B). Localized (Mondrian) was carried forward to test; global was the comparator, density-weighted a non-adopted check

$$c_i = \hat{y}_i \left(1 - \max(0,\ U_i - 0.20)\right)(1 - b), \tag{2}$$

where b is the registry's non-permanence buffer, applied unchanged. Interval quality and deductions are summarized in Table III and mapped per node in Fig. 3.

D. Evaluation protocol

Feature sets, model settings, the physics prior and the interval method were chosen by spatial five-fold cross-validation (CV) on training and validation nodes, using selection rules written and committed before each experiment. The test split was read once for all pre-specified models. After that read, one further model (the ensemble) was pre-specified on cross-validation alone and evaluated in a second, declared test read, and its test numbers are marked in Table II. Analyses that followed the first test read (generalization, two-date training, project aggregation, field cores, market data) used validation data or external data only and are reported as exploratory. Nine logged deviations and all rules are archived with the code.

IV. Experiments and Results

This section reports measurements only; their interpretation is deferred to Section V.

The three-way comparison at the earlier pilot's fixed weight ($\lambda = 0.3$, $w_{\text{nonneg}} = 0.1$) was re-run across five seeds $\{1, \ldots, 5\}$ on validation data.

A. Accuracy

Table II and Fig. 2(a) plots mean report accuracy. In R1 the ensemble reached test $R^2 = 0.277 \pm 0.035$ (mean $\pm$ one std. dev. for validation R^2 and prior-gen. correlation. Baseline GNN: R^2 mean = 0.005, std = 0.019; corr. mean = 0.544, std = 0.056. Physics-informed GNN: R^2 standard deviation (SD) over three seeds; mean = −0.040, std = 0.026; corr. mean = 0.632, std = 0.137. Linear regression is deterministic: R^2 = −0.030, corr. = 0.486 (no error bar; no seed dependence to show) . The λ sweep and NDWI/NDMI ablation (Section III-E/F)selected $\lambda = 0.1$ and the base feature set mechanically, by the rules stated where each is defined.

On the validation-internal calibration/check split(Section III-F), at 90 and 95% nominal respectively: global calibration reached 88.0 and 95.6% empirical coverage (mean width 307.6 and 367.8absolute error (MAE) 58.1 t/ha) ; localized (Mondrian) reached 92.2 and 95.6% (width 305.3 and 358.2 tC/ha); density-weighted reached 81.1 ha), against 0.203 for XGBoost and 0.170 for the physics-informed GNN, whose seed SD (0.054) was a quarter of the plain GNN's (0.236). A constant training mean scored −0.11, and the location-only model −0.79 despite a cross-validated R^2 of 0.42, so location alone did not transfer to unseen blocks. In R2 all non-linear models scored between 0.56 and 84.7% (width 220.2 0.60. On interior test nodes at least two hops from any other split, the ensemble kept $R^2 = 0.19$ (R1) and 241.5 t/ha), narrower but visibly undercovering both levels.

Table [removed] reports the single test evaluation (train 0.58 (R2). With all other settings fixed, correcting the image georeferencing alone raised the GNN's cross-validated R^2 : linear 0.236, baseline GNN 0.276, physics GNN 0.289; negative-prediction rate 0.0 for all three, not shownfrom 0.17 to 0.26, and adding the prior raised it from 0.39 to 0.41, within one standard error (0.03). The train ranking (physics > baseline > linear) reversed on test: the no-physics baseline GNN had the best test R^2. Edge incoherence (neighboring test-node jumps exceeding the 95th pct. of real training-side neighbor differences, threshold 123.12 t/ha) was 0.0 for both GNNs, near-zero for linear (graph message-passing generically, not the physics term)physics prior's correlation with the predictions it constrains was 0.14 on R1 test nodes, against 0.76 for the earlier proxy.

Table [removed] reports both methods on the single test evaluation. Global met or exceeded nominal coverage at both levels; localized (Mondrian)

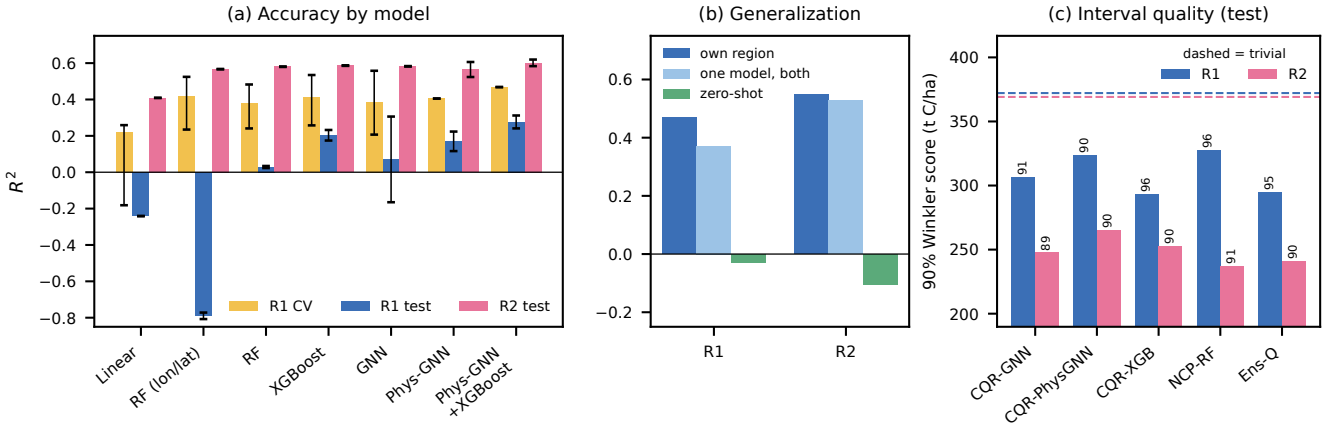


Fig. 2. (a) R^2 by model (R1 CV with fold range; test mean ± SD over 3 seeds). (b) Ensemble generalization. (c) 90% Winkler score on test; dashed: trivial interval; labels: coverage (%). CQR: conformalized quantile regression; NCP: normalized conformal; Ens-Q: quantile ensemble.

TABLE II
MODEL ACCURACY (R^2) AND FIELD-SITE ERROR

Model	R1 CV	R1 test	R2 test	Field (%)
Constant (train mean)	n/a	−0.11	−0.01	15.0
Linear	0.22	−0.24	0.41	n/a
Forest on lon/lat only	0.42	−0.79	0.57	n/a
Random forest	0.38	0.03	0.58	n/a
XGBoost	0.41	0.20	0.59	16.5
GNN	0.39	0.07	0.58	n/a
Physics-informed GNN	0.41	0.17	0.56	11.6
Ensemble (GNN+XGBoost)	**0.47**	**0.28**†	**0.60**†	**10.1**

Test: mean of 3 seeds. †Second, declared test read. Field: median absolute error at 25 core sites; published maps 14.1 and 15.1.

B. Generalization and stability

A single ensemble trained on both regions reached cross-validated $R^2 = 0.37$ (R1) and 0.53 (R2), ~~pre-registered (Section III-F)as the method to carry forward, undercovered more at both levels, despite matching/exceeding global there (Section IV-B).One test node (of 2,076)got an inverted (empty)interval under block-local correction at both levels, floored to zero width and excluded from Section VI's pricing examples; either way it counted as a coverage miss. Fig. 2(b) plots both against the $y = x$ perfect-calibration line~~against 0.47 and 0.55 for region-specific models, whereas models applied to a region they had not seen scored below zero (Fig. 2(b)). In that zero-shot setting, conformal intervals calibrated on the source region covered 76% to 89% of target validation nodes, and recalibrating them on half of the target's validation nodes restored 88% to 97%. Training the physics-informed GNN on both image dates raised the rank agreement of its predictions across dates from 0.44 to 0.88.

~~An earlier single-seed pilot found the physics-informed loss (Eq. [removed]) improved test R^2 over a no-physics baseline.The multi-seed check (Section IV-A) already showed this was fragile (physics-informed mean validation R^2 across 5 seeds (−0.040) sat below the baseline's (+0.005) even before the corrected split and pre-registered λ), and the final test evaluation confirms it: the no-physics baseline GNNhas the best test R^2 of the three. This is the actual finding, not reconciled with the earlier result by reframing; a single seed on a single split was not reliable here.~~

~~The physics-informed model's predictions correlate with the imposed mass-balance proxy more strongly than either non-physics model on every evaluation in this paper, multi-seed validation (0.632 vs. 0.544/0.486)and the single test evaluation (0.755~~

C. Intervals and crediting

Table III and Fig. 2(c) summarize the selected intervals (conformalized quantile GNN, global calibration). In R1 the intervals matched the trivial interval's width but scored 18% better on the Winkler score, and their width ranked absolute errors ($\rho = 0.54$, 95% confidence interval 0.51 to 0.58). In R2 they were 32% narrower than the trivial interval (191 vs. ~~0.682/0.686). This requires the same precise interpretation as before: the proxy correlated against is the exact quantity the physics loss trained toward, so this is a *prior-generalization* check (whether the constraint transfers to held-out nodes) , not independent physical validation, which this AOI's single-date,flux-tower-free data cannot supply. Whether this correlation reflects genuine physical behavior,not just successful optimization against its own training target, is this paper's most significant open question. Resolving it would require flux-tower or repeated field-plot carbon-flux measurements at this AOI, which do not currently exist, the clearest concrete next step for extending this work beyond a proxy-based prior. What changed under the corrected protocol is the effect's size, not its direction: a real but modest margin on test (0.069–0.073), not the 0.6-point gap the single-seed pilot reported. This is a direct product of~~ 282 t C/ha). Mondrian calibration covered 87.9% in R1, where only six validation blocks were available. Applied per node, the VM0033 rule withheld a median 12.7% (R1) and 2.1% (R2) of credits. Across all five interval methods, credited stock stayed within 0.84 to 0.90 (R1) and 0.95 to 0.98 (R2) of the buffer-only amount. As an illustrative standing-stock exposure, not credit issuance, the deducted share over the 2,076 R1 test nodes (about 4,700 ha) equals 0.71 million t CO_2e, or US$19.2 million at the ~~pre-registered protocol working as intended.~~ 2022 and 2023 average credit price (Section II). Credits are issued on stock changes, so this is an upper bound. Interval width did not track mangrove loss between 2010 and 2020 (Spearman −0.06 in both regions). Among the seven blue carbon projects with

TABLE III
INTERVALS (90%, TEST) AND CREDITING

	R1	R2
Coverage	91.5%	89.1%
Mean width / trivial (t C/ha)	273 / 274	191 / 282
Winkler score / trivial	306 / 372	247 / 369
ρ(width, abs. error)	0.54	0.15
Median VM0033 deduction per node	12.7%	2.1%
Project coverage, independent errors*	18%	33%
Project cov., bound / empirical*	91 / 91%	100 / 91%

*Realized errors of 2.5 km pseudo-projects, validation data.

issued credits (6.62 million), the median buffer was 11.0% and 73.5% of credits had been retired (registry records, Section II).

~~Section III-F pre-registered localized (Mondrian)CQR from validation-internal evidence, before test was read. On test it undercovered more than global at both levels (83.4%vs. 90% nominal; 92.4%vs. 95%, against global's 91.1%/96.4~~

When node uncertainties were aggregated to 2.5 km pseudo-projects as VM0033 prescribes (root-sum-of-squares, independent errors), the stated project uncertainty covered the realized aggregate error in only 2 of 11 projects in R1 (18%, exact 95% interval 2–52%) and 15 of 45 in R2 (33%, 20–49%). It implied no deduction. A fully correlated bound covered 91% and 100% (median deductions 14.6% and 3.1%)~~, likely because only 6 spatial superblocks had enough validation nodes ($\geq$ 30)for a stable local correction. This matches theory: [20] shows that local conformal validity needs calibration points dense enough within each local region, an *infill asymptotic* requirement this AOI's validation split plausibly did not satisfy, a density limit, not evidence against localization. Nor is the pattern isolated to this paper: applying Mondrian calibration to real-estate valuation, Hjort~~, and an empirical project-level conformal calibration on held-out pseudo-projects covered 91% in both regions.

D. Field soil cores

The 87 mangrove cores of the Coastal Carbon Library [29] were converted to stocks over the top 100 cm, using a quadratic organic-matter conversion of the form in Craft et al. ~~report "little to no improvement" over global in one scenario, cautioning that an apparent gain can instead reflect a non-conformity score failing to capture spatial trends. Localization gains in asset-pricing conformal calibration are setting-dependent, not automatic. global (simpler, the only method meeting nominal coverage on test) is defensible for near-term deployment; Section VI's pricing PoC uses global intervals accordingly. Both methods still produce test-set widths (318–383~~ [30] fitted on 679 regional samples ($R^2 = 0.91$), and matched to nodes within 300 ~~t/ha) several times the label's std. dev. (87.6~~m. At the 25 sites, the median absolute error was 10.1% for the ensemble, 11.6% for the physics-informed GNN, 16.5% for XGBoost, 14.1% and 15.1% for the Maxwell and Radabaugh maps, and 15.0% for a constant. Paired bootstrap 95% intervals of these differences included zero (ensemble minus Maxwell map -4.0 points, -8.1 to 8.2), and the Maxwell map ranked sites best (Spearman 0.63; ensemble 0.29). For the 13 cores sampled to at least 50 ~~t/ha), wide, not falsely confident.~~ cm, field-calibrated intervals covered 85% to 92% under leave-one-study-out and implied VM0033 deductions of 22% to 29%. The label product's own 95%

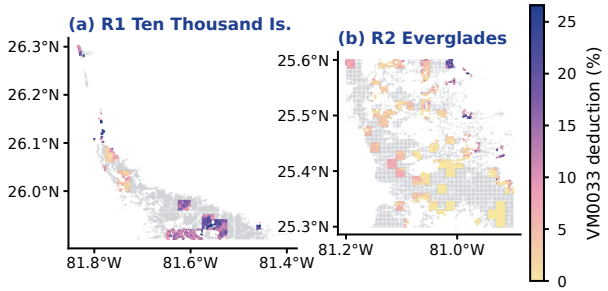


Fig. 3. Per-node VM0033 deduction (Eq. 2) on test nodes; gray: other splits.

prediction limits contained all 13 cores, but their median half-width equaled 101% of the estimate. Rescaled to 90% under a normal approximation, they would imply a VM0033 deduction of about 65% in both regions.

~~CQR interval width measures present-day estimation confidence, not non-permanence risk (future carbon loss to erosion, disturbance, sea-level rise) – the quantity buffer pools price; this paper does not establish the two correlate. We illustrate, without validating, how per-node estimation confidence could sit alongside, not replace, a non-permanence buffer, built on *global* CQR intervals per Section V's recommendation, not the localized (Mondrian) variant it cautions against. A relative interval width is the 90% -CQR (global)width divided by the test set's own median (341.1 t/ha; range 143.9–552.5 t/ha). A price multiplier is $1 - \alpha$(relative width $- 1$), clipped to $[0.5, 1.5]$, with $\alpha = 1.0$ an arbitrary illustrative sensitivity, not fit to market outcome. Table [removed] shows five nodes against Plan Vivo V5.0's flat 20% buffer.~~

V. DISCUSSION

~~Concretely, an issuer selling from both the well- (min-width)and poorly-characterized (max-width)node above applies the identical 20% discount today; under a confidence-adjusted mechanism, the former prices at a 50% premium,~~ **The physics-informed GNN and tree ensembles are complementary.** Alone, the ~~latter at a 50% discount, a plausible implication, not a validated outcome. α and the clipping bounds are stated, arbitrary choices. The flat-buffer value is not arbitrary: it is Plan Vivo V5.0's stated buffer, used only as a comparison point (VM0033 instead scores a project-specific value via its own risk tool, Section I~~physics-informed GNN matched XGBoost in cross-validation (0.41 each), but their mean outperformed every other final candidate in cross-validation (0.47 and 0.55, against at most 0.42 and 0.53) and every model on test (Table II). The GNN adds neighborhood structure and the hydroperiod prior that the trees do not see, and it was the most stable graph model across seeds and, trained on both dates, across seasons. The physics prior now acts as a weak constraint rather than a target the model copies.

Estimation uncertainty is of the same order as the buffer the market already holds. The per-node deduction in R1 (12.7%) is similar in size to the realized median buffer of issuing blue carbon projects (11.0%), yet today it is not applied. The similarity is context, not validation, because the two cover different risks. Calibration is what makes the deduction usable, since the label map's uncalibrated limits would withhold about five times as much in R1 (Section IV-D). Fig. ~~[removed] maps this interval width across the full test set, the spatial~~

~~pattern behind the location-differentiated pricing Table [removed] illustrates.~~ 3 shows that the deduction concentrates in specific areas, so it is a location-dependent quantity that a flat buffer cannot represent. Because width does not follow observed mangrove loss, the two discounts measure different risks and belong in separate factors, as in Eq. 2. **Business implication:** if validated on registry project data, a registry could adopt the deduction without changing its buffer policy, and an issuer can see in advance which areas carry the largest deduction and where additional field sampling would raise the credited volume most.

Aggregation under independent errors is the weak point of the registry rule. Spatial model errors are correlated, so root-sum-of-squares aggregation shrinks project uncertainty and grants no deduction even where realized errors exceed the allowance. Both a fully correlated bound and an empirical project-level calibration restored nominal coverage. Model-based crediting therefore needs an aggregation rule that accounts for spatial dependence.

What the intervals can and cannot do. Interval width is bounded by accuracy: with test $R^2 = 0.07$ for the GNN underlying the intervals in R1, at most about 4% narrowing over the trivial interval is attainable, and in R2 ($R^2 = 0.58$) the 32% achieved is close to the 35% bound. The value in R1 lies in adaptivity and Winkler score, not sharpness. Field-calibrated deductions exceed label-calibrated ones, a reminder that the label product carries its own error, in line with the departure of soil carbon maps from core data reported for United States tidal wetlands.

~~constrains the physics term to a cross-sectional relationship, not an actual dC/dt (not invalidating the prior-generalization result, defined relative to that same proxy). exists for this AOI; accretion/export are proxies (Section III-E)~~

VI. Limitations

Labels are a modeled product: the training label is a global map that disagrees with a field-calibrated regional map at node level (correlation -0.10 in R1). **Accuracy in R1 is moderate** ($R^2 = 0.28$). ~~, not a tidal-energy layer. (Sections III-F, V): it measures transfer, not physical correctness. , a disclosed scope choice. (Section IV-C, V), a data-density limit here, not spatial calibration generally [4], [20]. ; the five-seed check addresses seed variance, not regional generalization. , with an unfit sensitivity parameter and no market validation. (9.7% of nodes, 14.2% of test nodes, are 1-hop neighbors across a split); restricting Table [removed] to the 1,782 non-boundary test nodes gives lower R^2 for all three models (linear −0.504, baseline GNN −0.447, physics GNN −0.494) but the same ranking, so reported numbers carry a small, non-model-specific optimistic bias from this leakage. remain unaddressed~~**The ensemble was evaluated in a second test read,** declared and limited to one pre-specified model. **Single tide gauges** represent each region, and R2's is about 70 km away. **Field cores are sparse** (13 deep enough for 100 cm stocks) and may overlap the global map's training data. **Market outcomes** were not modeled.

VII. Conclusion

~~A pre-registered, two-stage protocol (validation-only selection, single held-out test confirmation) caught two overstated results a single run would have missed: an earlier single-seed pilot's headline finding, and a theoretically-motivated localized calibration refinement favorable on validation-internal evidence alone. What survives (the~~ Per-node uncertainty from a physics-informed ~~model's smaller but real prior-generalization advantage, and honestly wide, node-differentiated CQR intervals) are two separate results, not one flowing from the other: the CQR pipeline priced here used data loss only, without the physics term. Together they motivate the market-facing contribution: an estimation-confidence signal that could complement, not replace, flat buffer pools, deployed here on the simpler, better-calibrated global variant. Future work : a second Sentinel-2 date, a higher-volume or second-region localized-conformal test, deep-ensemble/MC-dropout comparisons, and market-outcome validation of the pricing mechanism against non-permanence risk~~GNN and XGBoost ensemble, calibrated on held-out map labels, can supply the VM0033 uncertainty deduction. The deduction (median 12.7% in R1) is of the same order as the 11.0% buffers that issuing projects hold, with interval coverage of 89% to 92% in two regions. The methodology's independent-error aggregation covered only 18% to 33% of realized project errors, and against 87 field cores the ensemble had the lowest site-level median error (10.1%), although the margin over published maps was within sampling uncertainty. Future work will test a correlation-aware aggregation with registry project data, add regions with denser field cores, and replace the steady-state prior with repeated elevation and accretion measurements.

Code Availability

Code, data pipeline, selection rules and results are available at https://github.com/Nahla-Nabil/PIGNN-UQ.

References

[1] Verra, "VM0033: Methodology for tidal wetland and seagrass restoration, v2.1," https://verra.org/methodologies/vm0033-methodology-for-tidal-wetland-and-seagrass-restoration-v2-1/, 2023.

[2] K. R. Radabaugh *et al.*, "A spatial model comparing above- and belowground blue carbon stocks in southwest Florida mangroves and salt marshes," *Estuaries and Coasts*, vol. 46, pp. 1536–1556, 2023, doi: 10.1007/s12237-023-01217-7.

[3] G. Singh, G. Moncrieff, Z. Venter, K. Cawse-Nicholson, J. Slingsby, and T. B. Robinson, "Uncertainty quantification for probabilistic machine learning in earth observation using conformal prediction," *Scientific Reports*, vol. 14, p. 16166, 2024, doi: 10.1038/s41598-024-65954-w.

[4] X. Lou, P. Luo, and L. Meng, "GeoConformal prediction: A model-agnostic framework for measuring the uncertainty of spatial prediction," *Annals of the American Association of Geographers*, vol. 115, no. 8, pp. 1971–1998, 2025, doi: 10.1080/24694452.2025.2516091.

[5] T. L. Maxwell *et al.*, "Global mangrove soil organic carbon stocks dataset at 30 m resolution for the year 2020 based on spatiotemporal predictive machine learning," *Data in Brief*, vol. 50, p. 109621, 2023, doi: 10.1016/j.dib.2023.109621.

[6] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019, doi: 10.1016/j.jcp.2018.10.045.

[7] M. Taghizadeh, Z. Zandsalimi, M. A. Nabian, M. Shafiee-Jood, and N. Alemazkoor, "Interpretable physics-informed graph neural networks for flood forecasting," *Computer-Aided Civil and Infrastructure Engineering*, vol. 40, pp. 2629–2649, 2025, doi: 10.1111/mice.13484.

[8] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Computing Surveys*, vol. 55, no. 4, pp. 1–37, 2023, doi: 10.1145/3514228.

[9] W. L. Hamilton, Z. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, arXiv:1706.02216.

[10] A. Dutta Roy *et al.*, "Remote sensing-based mangrove blue carbon assessment in the Asia-Pacific: A systematic review," *Science of The Total Environment*, vol. 938, p. 173270, 2024, doi: 10.1016/j.scitotenv.2024.173270.

[11] J. Sanderman *et al.*, "A global map of mangrove forest soil carbon at 30 m spatial resolution," *Environmental Research Letters*, vol. 13, p. 055002, 2018, doi: 10.1088/1748-9326/aabe1c.

[12] A. S. Rovai *et al.*, "Global controls on carbon storage in mangrove soils," *Nature Climate Change*, vol. 8, pp. 534–538, 2018, doi: 10.1038/s41558-018-0162-5.

[13] K. W. Krauss *et al.*, "How mangrove forests adjust to rising sea level," *New Phytologist*, vol. 202, pp. 19–34, 2014, doi: 10.1111/nph.12605.

[14] J. R. Holmquist *et al.*, "Accuracy and precision of tidal wetland soil carbon mapping in the conterminous United States," *Scientific Reports*, vol. 8, p. 9478, 2018, doi: 10.1038/s41598-018-26948-7.

[15] D. R. Roberts *et al.*, "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, pp. 913–929, 2017, doi: 10.1111/ecog.02881.

[16] P. Ploton *et al.*, "Spatial validation reveals poor predictive performance of large-scale ecological mapping models," *Nature Communications*, vol. 11, p. 4540, 2020, doi: 10.1038/s41467-020-18321-y.

[17] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, arXiv:1905.03222.

[18] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, and L. Wasserman, "Distribution-free predictive inference for regression," *Journal of the American Statistical Association*, vol. 113, pp. 1094–1111, 2018, doi: 10.1080/01621459.2017.1307116.

[19] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005, doi: 10.1007/b106715.

[20] H. Mao, R. Martin, and B. J. Reich, "Valid model-free spatial prediction," *Journal of the American Statistical Association*, vol. 119, no. 546, pp. 904–914, 2024, doi: 10.1080/01621459.2022.2147531.

[21] R. L. Winkler, "A decision-theoretic approach to interval estimation," *Journal of the American Statistical Association*, vol. 67, pp. 187–191, 1972, doi: 10.1080/01621459.1972.10481224.

[22] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *Journal of the American Statistical Association*, vol. 102, pp. 359–378, 2007, doi: 10.1198/016214506000001437.

[23] S. Farahmand, N. Hilmi, and C. M. Duarte, "The rise and flows of blue carbon credits advance global climate and biodiversity goals," *npj Ocean Sustainability*, vol. 4, p. 39, 2025, doi: 10.1038/s44183-025-00141-6.

[24] P. Quartson *et al.*, "Voluntary Registry Offsets Database v2026-08," https://gspp.berkeley.edu/berkeley-carbon-trading-project/offsets-database, 2026, Berkeley Carbon Trading Project, University of California, Berkeley.

[25] P. Bunting *et al.*, "Global mangrove extent change 1996–2020: Global Mangrove Watch version 3.0," *Remote Sensing*, vol. 14, p. 3657, 2022, doi: 10.3390/rs14153657.

[26] M. Drusch *et al.*, "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sensing of Environment*, vol. 120, pp. 25–36, 2012, doi: 10.1016/j.rse.2011.11.026.

[27] L. Breiman, "Random forests," *Machine Learning*, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[28] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.

[29] Coastal Carbon Network, "Database: Coastal Carbon Library (version 1.7.0)," Smithsonian Environmental Research Center, dataset, 2025, doi: 10.25573/serc.21565671, accessed 2026-10-09.

[30] C. B. Craft, E. D. Seneca, and S. W. Broome, "Loss on ignition and Kjeldahl digestion for estimating organic carbon and total nitrogen in estuarine marsh soils: Calibration with dry combustion," *Estuaries*, vol. 14, pp. 175–179, 1991, doi: 10.2307/1351691.